World Climate Research Programme · Fresh Eyes on CMIP

Climate Data Analysis Tools and Tutorials Workshop

Welcome & Introduction to Fresh Eyes on CMIP

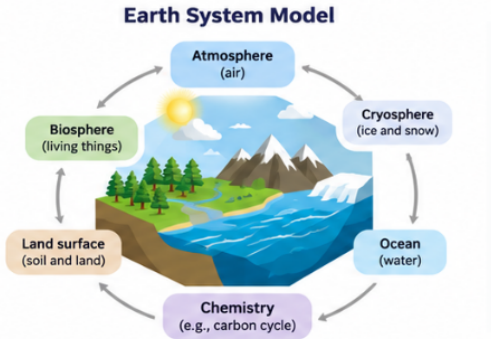
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fresh-eyes.wcrp-cmip.org

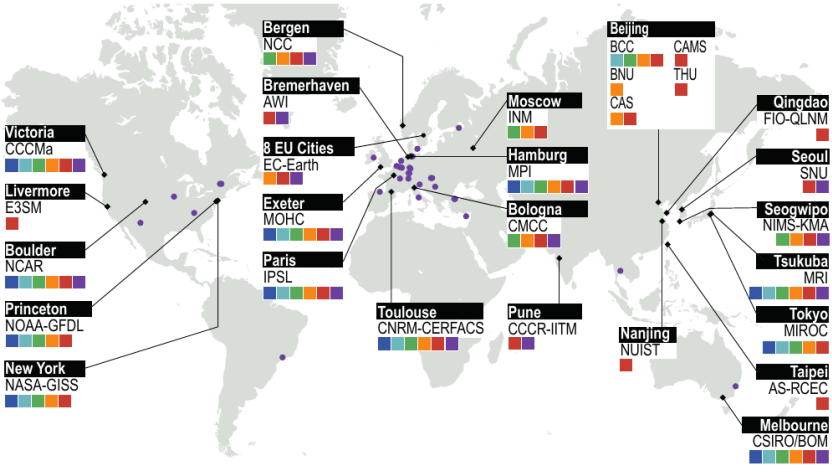
What is Coupled Model Intercomparison Project (CMIP)?

- CMIP is an international collaboration that brings together climate modelling groups to conduct and compare climate model experiments.
- It is coordinated by the **World Climate Research Programme (WCRP).**
- It supports climate research and **IPCC assessments.**



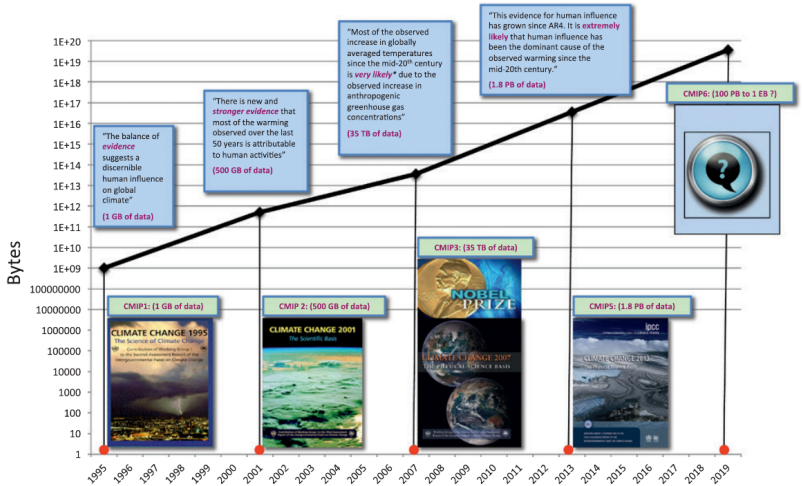


CMIP – A global modelling community



Source: IPCC AR6

The growing volume of CMIP data



Source: BAMS (2016)

What is Fresh Eyes on CMIP?

An initiative that engages primarily **early-career researchers** in the CMIP community.

Fresh perspectives are welcome — being "new to CMIP" is not necessarily about age.

- **Connect** — Build networks and share experiences.
- **Learn** — Improve access to CMIP knowledge and resources.
- **Contribute** — Bring fresh perspectives to community projects.



https://fresh-eyes.wcrp-cmip.org/

Aggregating CMIP data analysis tools in a common Github repository

Create a structured, functional, tested & documented Github repository of CMIP data analysis tools contributed by members of the Fresh Eyes on CMIP community.

Compiling existing CMIP/climate data analysis tools & tutorials

Update the existing CMIP list of tools & create a collection of existing tutorials.

Entry level documentation for first time CMIP users

Develop a website collating useful information for first time CMIP users.

Fresh Eyes platform

Provide & maintain an online platform for FEoC members to develop networks, facilitate projects, foster collaboration & enable knowledge exchange.

Fresh Eyes workshop

Foster collaboration between global early career researchers & the broader CMIP community through virtual workshops.

Developing guidelines for working with CMIP multi-model ensembles

Review existing literature to develop guidelines on how to best work with CMIP multi-model ensembles.

Responsible use of CMIP data

Determine the practices behind a responsible use of CMIP data, accounting for model limitations & the diverse existing downstream applications. Collaborate with the Regional Information for Society WCRP activity to disseminate these practices.



Evaluating CMIP6Plus & CMIP7 forcing datasets

Evaluate the forcing datasets before their public release & document the evaluation results in conjunction with the release of the CMIP7 forcing datasets.

Consistency between paleo & modern forcing

Assess the consistency between paleo forcing & modern forcing datasets & evaluate the impacts for CMIP7 experiments.

Identifying & communicating model biases

Synthesize literature & community sources to document climate model biases & communicate possible solutions to mitigate these biases.

Supporting CMIP7 data request

Collaborate with the Data Request Task Team to provide a consistent, robust & user-oriented set of parameters on which to build the CMIP7 archive.

CMIP6 data usage report

Comprehensive statistical analysis of usage of CMIP6 data by the community.

Current Projects

Interested in joining Fresh Eyes on CMIP?



Collecting existing CMIP/climate data analysis tools and tutorials

From existing community resources to easier discovery and use...

Project objectives:

1. **Upgrade** the WCRP CMIP catalogue of climate-data analysis tools.
2. **Collect** existing tutorials covering climate-data analysis and climate science.
3. **Organize** an interactive online workshop to showcase available tools and tutorials including practical examples.



https://wcrp-cmip.org/tools/

CMIP Coupled Model Intercomparison Project — TUTORIALS

Tutorials for using climate data

A WCRP project — World Climate Research Programme

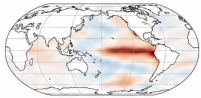


Climate Unboxed

COMET MedEd

RESEARCH COMPUTING IN EARTH SCIENCES
Fall 2017 | Columbia University Department of Earth and Environmental Sciences

Research Computing in Earth Sciences



Earth and Environmental Data Science

Climatematch Academy

summer-school-2022 Public

ISENES Summer School 2022

Quantus x Climate

ESMValTool
Earth System Model Evaluation Tool

ESMValTool Tutorials

AOGS2024
21ST ANNUAL MEETING
Pyeongchang, Gangwon-do
Home to Winter Olympics
23 - 28 Jun 2024
CMIP Coupled Model Intercomparison Project

CMIP AOGS 2024 Tutorials

https://wcrp-cmip.org/tutorials/

Help us build the community resource

- **Know a useful tool – or developed one yourself?**
  → Suggest an addition to the CMIP tools collection.

  https://wcrp-cmip.org/tools/

- **Know a useful tutorial – or created one yourself?**
  → Share it for consideration in the tutorial collection.

  https://wcrp-cmip.org/tutorials/

Your feedback can shape the collection.

Today: learn & discuss

Our goal is to make climate-data analysis resources more **discoverable, accessible and useful**.

Coming next:

- Demonstrations and hands-on examples
- Questions, exchange and collaboration

Thank you – and enjoy the workshop! ☺